

Notes: Adão, Kolesár, and Morales (QJE, 2019)
Shift-Share Designs: Theory and Inference

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1 Big picture

This paper studies a methodological problem that appears in many empirical economics papers: **how should researchers do inference in shift-share, or Bartik-style, research designs?** A shift-share design regresses a regional outcome on a weighted average of sector-level shocks. The weights are typically predetermined local exposure shares.

The canonical regression is

$$Y_i = \beta X_i + Z_i' \delta + \varepsilon_i, \quad X_i = \sum_{s=1}^S w_{is} X_s, \quad (1)$$

where Y_i is a regional outcome, X_s is a sector-level shifter, and w_{is} is region i 's exposure to sector s .

The paper's main point is simple but important: **ordinary robust standard errors and geographically clustered standard errors can be badly wrong in shift-share designs.** They miss the fact that regions with similar exposure shares have similar values of the shift-share regressor and may also have similar residuals, even if they are geographically far apart.

The authors show this problem with a placebo exercise. They generate random sectoral shocks and combine them with actual U.S. commuting-zone employment shares. Since the shocks are random, the true effect is zero. Yet standard inference rejects the zero-effect null far too often. The paper then derives new inference procedures, now commonly called **AKM** and **AKM0**, that are valid under arbitrary cross-region residual correlation when the identifying variation comes from many independent sector-level shifters.

2 Incremental contribution of the paper

The paper's incremental contribution is the inference theory for a class of empirical designs that had already become common before the inference problem was fully understood.

1. **Diagnoses a concrete inference failure.** The paper shows that familiar standard errors can produce rejection rates as high as roughly 55% in placebo samples where the true rejection rate should be 5%.
2. **Explains why the failure occurs.** A stylized economic model shows that regional outcomes can depend on both observed and unobserved sector-level shocks through multiple shift-share terms. Therefore, residuals are naturally correlated across regions with similar shares.
3. **Changes the level of inference.** Rather than clustering by geography, the paper treats the sector-level shifters as the random objects and conditions on shares, controls, and residuals.
4. **Allows arbitrary regional residual correlation.** Because inference is based on random assignment of shifters, the method does not need to specify a spatial covariance structure for regional residuals.

5. **Provides usable formulas.** The paper derives practical confidence intervals and standard errors for OLS and IV shift-share regressions, including versions that allow correlated shifters within sector clusters.
6. **Revisits influential applications.** The authors apply the method to Chinese import competition and Bartik labor-supply regressions, showing when conclusions are robust and when uncertainty is meaningfully larger.

3 Data and measurement

3.1 Placebo data

The placebo exercise uses actual U.S. local labor market data but fake sectoral shocks. The regional units are **722 commuting zones**. The sectors are **396 four-digit SIC manufacturing industries**. The shares are 1990 employment shares from County Business Patterns. Outcomes are 2000–2007 changes in employment rates and average log weekly wages, following the data construction in Autor, Dorn, and Hanson-style local labor market studies.

For each of 30,000 placebo samples, the authors draw independent random sector-level shifters and construct

$$X_i^m = \sum_s w_{is} X_s^m. \quad (2)$$

The true coefficient is zero because the shifters are randomly generated and independent of actual labor market outcomes.

3.2 Empirical applications

The first application revisits the effect of Chinese import competition on U.S. commuting zones. The paper studies the same broad reduced-form and IV specifications used in Autor, Dorn, and Hanson (2013), comparing conventional confidence intervals with the new AKM and AKM0 intervals.

The second application revisits Bartik-style estimates of the regional inverse labor supply elasticity. The authors estimate first-stage, reduced-form, and 2SLS equations using different shift-share instruments, including the standard Bartik instrument, a leave-one-out version, and the ADH China-shock instrument.

4 Models and empirical specifications

4.1 The shift-share regression

The core empirical object is

$$Y_i = \beta X_i + Z_i' \delta + \varepsilon_i, \quad X_i = \sum_s w_{is} X_s. \quad (3)$$

The empirical issue is not how to estimate β by OLS, but how to estimate the sampling uncertainty of $\hat{\beta}$ when X_i is mechanically correlated across regions through common sectoral shifters.

4.2 Economic model behind residual correlation

The paper introduces a stylized regional labor market model. Regional employment and wages depend on sector-level labor-demand shocks and group-specific labor-supply shocks. A first-order approximation delivers equations of the form

$$\hat{L}_i = \sum_s l_{is}^0 (\theta_{is} \hat{\chi}_s + \lambda_i \hat{\mu}_s + \lambda_i \hat{\eta}_{is}) + (1 - \lambda_i) \left(\sum_g \tilde{w}_{ig} \hat{\nu}_g + \hat{\nu}_i \right), \quad (4)$$

where the observed shifters may be $\hat{\chi}_s$, but unobserved terms such as $\hat{\mu}_s$ and $\hat{\nu}_g$ can also enter outcomes through shift-share structures.

The economic message is that a regression residual may contain unobserved shift-share components. Thus two regions with similar sector shares can have correlated residuals, even if they are in different states.

4.3 Potential-outcome framework

The paper writes the potential outcome as

$$Y_i(x_1, \dots, x_S) = Y_i(0) + \sum_{s=1}^S w_{is} x_s \beta_{is}. \quad (5)$$

Here $w_{is} \beta_{is}$ is the effect on region i of a one-unit shock to sector s . The treatment effects β_{is} can be heterogeneous across sectors and regions.

With no controls, the estimand is

$$\beta = \frac{\sum_i E[X_i Y_i \mid \mathcal{F}_0]}{\sum_i E[X_i^2 \mid \mathcal{F}_0]}, \quad (6)$$

where $\mathcal{F}_0$ contains the fixed shares, potential outcomes, and heterogeneous effects. Under the paper's assumptions, this equals a weighted average of the heterogeneous effects:

$$\beta = \frac{\sum_i \sum_s \pi_{is} \beta_{is}}{\sum_i \sum_s \pi_{is}}, \quad \pi_{is} = w_{is}^2 \text{Var}(X_s \mid \mathcal{F}_0). \quad (7)$$

4.4 AKM standard error intuition

A simplified version of the AKM variance formula can be understood as aggregating residuals to the sector-shock level. Define

$$\hat{R}_s = \sum_i w_{is} \hat{\epsilon}_i. \quad (8)$$

Then the middle piece of the variance estimator has the form

$$\hat{V}^{AKM} = \sum_s X_s^2 \hat{R}_s^2. \quad (9)$$

With controls, the paper residualizes the relevant objects. With clustered shifters, the sector-level terms are aggregated within shifter clusters.

The intuition is the same as cluster-robust inference: if treatment varies at the sector-shock level, inference should aggregate the residual variation to that level. The difference is that each region is usually exposed to many sectors, so the sector-level residual is a share-weighted sum of regional residuals.

5 Identification and inference design

The paper’s identifying assumption is about the **shifters**, not the shares. Conditional on shares and controls, the shifters are assumed to be as good as randomly assigned and independent across sectors or across sector clusters.

The baseline assumptions are:

1. **Random shifters:** $E[X_s \mid \mathcal{F}_0] = 0$.
2. **Independent shifters:** X_s are independent across sectors, or independent across specified sector clusters.
3. **Many relevant sectors:** no single sector dominates the shift-share variation.
4. **Fixed shares for inference:** the realized shares are conditioned on, so inference does not require shares to be randomly assigned.

This framework lets the residuals ε_i have arbitrary spatial, industrial, or other correlation across regions. That is the key benefit of the approach. The cost is that the researcher must justify independence, or at least appropriate clustering, of the shifters.

6 Main results

6.1 Traditional inference overrejects

In the placebo exercise, robust and state-clustered standard errors are much smaller than the true sampling dispersion of the OLS estimates. For example, when the outcome is the change in the employment rate, robust standard errors reject the zero-effect null in 48.5% of placebo samples and state-clustered standard errors reject in 38.1%. For manufacturing employment, the rejection rates are even higher: 55.7% and 44.8%.

The interpretation is direct: conventional inference mistakes share-induced residual correlation for independent information.

6.2 AKM and AKM0 fix the placebo problem

The new AKM and AKM0 procedures produce standard errors much closer to the true dispersion in placebo estimates. AKM is slightly liberal in the baseline placebo, with rejection rates around 7.5–9.1%. AKM0 is more conservative and closer to exact size, with rejection rates around 4.3–4.5%.

6.3 Finite-sample issues still matter

The placebo extensions show that inference quality depends on the effective number and balance of shifters. When the number of sectors falls, or when a large sector dominates the shift-share regressor, all methods can perform worse. In the table with a large sector in the shift-share regressor, even AKM and AKM0 overreject heavily.

6.4 Correlated shifters require appropriate shifter clustering

When shifters are correlated within sector groups, assuming independent sector shocks is not enough. The paper shows that AKM/AKM0 can work when the researcher clusters shifters at a sufficiently broad sector level. If shocks are correlated within one-digit sectors but the researcher only clusters at two-digit sectors, confidence intervals can remain too narrow.

6.5 Applications change some confidence intervals

For the China import competition application, AKM and AKM0 confidence intervals are wider than robust or state-clustered confidence intervals. The import-competition estimates remain economically large, but the new intervals show that conventional intervals can overstate precision.

For the Bartik labor-supply application, the standard Bartik IV yields confidence intervals close to conventional methods. In contrast, the ADH IV generates substantially wider AKM/AKM0 first-stage and reduced-form intervals. The difference suggests that the importance of shift-share residual structure varies across applications.

7 Tables and figures

7.1 Table I. Conventional robust and state-clustered placebo inference and Table II. AKM and AKM0 placebo inference

TABLE I
STANDARD ERRORS AND REJECTION RATE OF THE HYPOTHESIS $H_0: \beta = 0$ AT 5% SIGNIFICANCE LEVEL

	Estimate		Median std. error		Rejection rate	
	Mean	Std. dev.	Robust	Cluster	Robust	Cluster
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: Change in the share of working-age population						
Employed	-0.01	2.00	0.73	0.92	48.5%	38.1%
Employed in manufacturing	-0.01	1.88	0.60	0.76	55.7%	44.8%
Employed in nonmanufacturing	0.00	0.94	0.58	0.67	23.2%	17.6%
Panel B: Change in average log weekly wage						
Employed	-0.03	2.66	1.01	1.33	47.3%	34.2%
Employed in manufacturing	-0.03	2.92	1.68	2.11	26.7%	16.8%
Employed in nonmanufacturing	-0.02	2.64	1.05	1.33	45.4%	33.7%

Notes: For the outcome variable indicated in the leftmost column, this table indicates the mean and standard deviation of the OLS estimates of β in equation (1) across the placebo samples (columns (1) and (2)), the median standard error estimates (columns (3) and (4)), and the percentage of placebo samples for which we reject the null hypothesis $H_0: \beta = 0$ using a 5% significance level test (columns (5) and (6)). Robust is the Eicker-Huber-White standard error, and Cluster is the standard error that clusters CEs in the same state. Results are based on 30,000 placebo samples.

Table I. Conventional robust and state-clustered placebo inference

TABLE II
MEDIAN STANDARD ERRORS AND REJECTION RATES FOR $H_0: \beta = 0$ AT 5% SIGNIFICANCE LEVEL

	Estimate		Median eff. s.e.		Rejection rate	
	Mean	Std. dev.	AKM	AKM0	AKM	AKM0
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: Change in the share of working-age population						
Employed	-0.01	2.00	1.90	2.21	7.8%	4.5%
Employed in manufacturing	-0.01	1.88	1.77	2.06	8.0%	4.3%
Employed in nonmanufacturing	0.00	0.94	0.89	1.04	8.2%	4.5%
Panel B: Change in average log weekly wage						
Employed	-0.03	2.66	2.57	2.99	7.5%	4.3%
Employed in manufacturing	-0.03	2.92	2.74	3.18	9.1%	4.5%
Employed in nonmanufacturing	-0.02	2.64	2.55	2.96	7.8%	4.5%

Notes: For the outcome variable indicated in the leftmost column, this table indicates the mean and standard deviation of the OLS estimates of β in equation (1) across the placebo samples (columns (1) and (2)), the median effective standard error estimates (columns (3) and (4)), and the percentage of placebo samples for which we reject the null hypothesis $H_0: \beta = 0$ using a 5% significance level test (columns (5) and (6)). AKM is the standard error in Remark 5, and AKM0 is the confidence interval in Remark 6. The median effective standard error is equal to the median length of the corresponding 95% confidence interval divided by 2×1.96 . Results are based on 30,000 placebo samples.

Table II. AKM and AKM0 placebo inference

Read: Table I shows the failure of conventional inference. The mean placebo estimates are approximately zero, as they should be, but robust and state-clustered standard errors are far below the empirical standard deviation of the estimates. The result is massive overrejection. Table II repeats the exercise using AKM and AKM0. The AKM standard errors closely track the empirical dispersion, and AKM0 has rejection rates very close to the nominal 5% level.

Economic message: The econometric problem is not small-sample noise. It is a structural feature of shift-share designs: observations with similar exposure shares are not independent pieces of evidence. Correct inference must respect the shifter-level source of variation.

7.2 Table III. Sensitivity to number of sectors, shifter distributions, and residual structure and Table IV. Correlation in sectoral shocks

TABLE III
ALTERNATIVE NUMBER OF SECTORS, SHIFTER DISTRIBUTIONS, AND RESIDUALS' CORRELATION PATTERNS

	Estimate		Median eff. s.e.			Rejection rate		
	Mean	Std. dev.	Robust	Cluster	AKM	Robust	Cluster	AKM
	(1)	(2)	(3)	(4)	(5)	(7)	(8)	(10)
Panel A: Sensitivity to the number of sectors								
2-digit ($S = 20$)	-0.01	3.19	0.65	0.96	6.06	70.8%	56.1%	12.0%
3-digit ($S = 100$)	0.00	2.25	0.73	0.94	2.18	54.2%	42.5%	7.5%
Panel B: Sensitivity to the variance of the shifters								
$\sigma^2 = 0.5$	-0.04	6.33	2.33	2.91	6.04	48.5%	38.0%	7.9%
$\sigma^2 = 10$	-0.00	1.41	0.52	0.65	1.25	37.8%	27.8%	7.5%
Panel C: Log-normal shifters								
$\sigma^2 = 5$	0.27	2.26	0.86	1.05	2.17	3.7%	35.3%	7.7%
Panel D: Heteroskedastic shifters								
$\lambda = 3$	-0.01	1.63	0.55	0.72	1.51	52.1%	40.1%	8.7%
$\lambda = 7$	0.01	1.38	0.44	0.58	1.23	53.7%	41.1%	9.3%
Panel E: Simulated state-level shocks in regression residual								
0.00	2.11	0.86	1.11	1.99	2.32	42.8%	30.4%	7.9%
Panel F: Simulated "large" sector shifter in regression residual								
-0.01	2.01	0.74	0.92	1.90	2.21	48.4%	37.8%	7.9%
Panel G: Including a "large" sector in shift-share regressor								
-0.02	4.25	0.59	0.76	1.18	1.34	92.0%	89.6%	77.2%

Notes: All estimates in this table use the change in the share of the working-age population used in each CE as the outcome variable Y_t in equation (1). This table indicates the mean and standard deviation of the OLS estimates of β in equation (1) across the placebo samples (columns (1) and (2)), the median effective standard error estimates (columns (3) to (5)), and the percentage of placebo samples for which we reject the null hypothesis $H_0: \beta = 0$ using a 5% significance level test (columns (7) to (10)). Robust is the Eicker-Huber-White standard error, Cluster is the standard error that clusters CEs in the same state, AKM is the standard error in Remark 5, and AKM0 is the confidence interval in Remark 6. For each inference procedure, the median effective standard error is equal to the median length of the corresponding 95% confidence interval divided by 2×1.96 . Results are based on 30,000 placebo samples. This table presents results for placebo simulations that depart from the baseline; the results should thus be compared to those in Tables I and II. In Panel A, we reduce the number of sectors relative to the baseline. In Panel B, we change the variance of the distribution from which all shifters are drawn. In Panel C, we assume that the distribution from which all shifters are drawn is log-normal (recentered at 0) with variance equal to 5. In Panel D, we allow the variance of the shock in each sector to be heteroskedastic, $\sigma^2 = 5 + \lambda \cdot (n_i - \bar{n})$. In Panel E, we simulate state-level shocks and include them in our regression residual. In Panels F and G, we simulate

Table III. Sensitivity to number of sectors, shifter distributions, and residual structure

TABLE IV
CORRELATION IN SECTORAL SHOCKS

	Estimate		Median eff. s.e.				Rejection rate			
			Independent		Two-digit SIC		Independent		Two-digit SIC	
	Mean	Std. dev.	Robust	Cluster	AKM	AKM0	Robust	Cluster	AKM	AKM0
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Panel A: Simulated shifters with correlation within three-digit SIC sectors										
$\rho = 0.00$	-0.01	2.00	0.74	0.92	1.91	2.22	1.86	2.64	48.6%	37.8%
$\rho = 0.50$	0.02	2.14	0.77	1.07	2.03	2.16	2.34	2.87	49.2%	32.3%
$\rho = 1.00$	0.01	2.27	0.76	1.08	1.99	2.10	2.38	3.14	52.2%	35.0%
Panel B: Simulated shifters with correlation within two-digit SIC sectors										
$\rho = 0.00$	-0.01	1.99	0.73	0.92	1.90	2.22	1.86	2.65	48.2%	37.7%
$\rho = 0.50$	-0.01	2.73	0.73	1.13	1.82	1.89	2.92	4.07	62.3%	43.2%
$\rho = 1.00$	0.01	3.20	0.69	1.18	1.67	1.63	3.38	6.16	68.4%	48.1%
Panel C: Simulated shifters with correlation within one-digit SIC sectors										
$\rho = 0.00$	0.01	1.98	0.73	0.92	1.90	2.21	1.85	2.65	48.5%	37.7%
$\rho = 0.50$	0.02	4.95	0.74	1.41	1.88	1.59	2.42	2.36	84.2%	72.3%
$\rho = 1.00$	0.42	59.63	0.71	1.74	1.86	1.02	2.94	1.67	90.2%	78.3%

Notes: All estimates in this table use the change in the share of the working-age population employed in each CE as the outcome variable Y_t in equation (1). This table indicates the mean and standard deviation of the OLS estimates of β in equation (1) across the placebo samples (columns (1) and (2)), the median effective standard error estimates (columns (3) to (6)), and the percentage of placebo samples for which we reject the null hypothesis $H_0: \beta = 0$ using a 5% significance level test (columns (7) to (10)). Robust is the Eicker-Huber-White standard error, Cluster is the standard error that clusters CEs in the same state, in columns (3) and (4), AKM is the standard error in Remark 5, in columns (5) and (6), and AKM0 is the confidence interval in Remark 6 as indicated in Section VA. For each inference procedure, the median effective standard error is equal to the median length of the corresponding 95% confidence interval divided by 2×1.96 . Results are based on 30,000 placebo samples.

Table IV. Correlation in sectoral shocks

Read: Table III shows that the method performs well in many placebo variants, but it also reveals important weak spots. Reducing the number of sectors worsens conventional overrejection, and including a dominant large sector makes inference difficult even for AKM-type methods. Table IV shows that when sectoral shocks are correlated, the researcher must cluster shifters at the level where correlation remains.

Economic message: AKM is not a magic correction that removes every problem. It is valid under an explicit design: many shocks, no dominant shifter, and independence across the chosen shifter clusters. The table tells the applied researcher what to diagnose before trusting a shift-share design.

7.3 Table V. Chinese import competition and U.S. commuting zones and Table VI. Inverse labor supply elasticity

TABLE V
EFFECT OF CHINESE EXPORTS ON U.S. COMMUTING ZONES (AUTOR, DORN, AND HANSON 2013)

	Change in the employment share			Change in avg. log weekly wage		
	All (1)	Manuf. (2)	Nonmanuf. (3)	All (4)	Manuf. (5)	Nonmanuf. (6)
Panel A: 2SLS Regression						
$\hat{\beta}$	-0.77					
Robust	[-1.10, -0.45]	[-0.78, -0.41]	[-0.47, 0.12]	[-1.28, -0.29]	[-0.81, 1.11]	[-1.27, -0.25]
Cluster	[-1.12, -0.42]	[-0.79, -0.40]	[-0.45, 0.10]	[-1.26, -0.26]	[-0.81, 1.11]	[-1.28, -0.24]
AKM	[-1.25, -0.30]	[-0.84, -0.35]	[-0.54, 0.18]	[-1.37, -0.15]	[-0.81, 1.11]	[-1.42, -0.10]
AKM0	[-1.62, -0.39]	[-1.01, -0.36]	[-0.84, 0.14]	[-1.77, -0.17]	[-1.49, 1.05]	[-1.97, -0.19]
Panel B: OLS reduced-form regression						
$\hat{\beta}$	-0.49	-0.38	-0.11	-0.48	0.10	-0.48
Robust	[-0.71, -0.27]	[-0.48, -0.28]	[-0.31, 0.08]	[-0.80, -0.16]	[-0.50, 0.69]	[-0.83, -0.13]
Cluster	[-0.64, -0.34]	[-0.45, -0.30]	[-0.27, 0.05]	[-0.78, -0.18]	[-0.51, 0.70]	[-0.81, -0.15]
AKM	[-0.81, -0.17]	[-0.52, -0.23]	[-0.35, 0.12]	[-0.88, -0.07]	[-0.50, 0.69]	[-0.93, -0.03]
AKM0	[-1.24, -0.24]	[-0.67, -0.25]	[-0.64, 0.08]	[-1.27, -0.10]	[-1.16, 0.61]	[-1.47, -0.11]
Panel C: 2SLS first-stage						
$\hat{\beta}$			0.63			
Robust			[0.46, 0.80]			
Cluster			[0.45, 0.81]			
AKM			[0.53, 0.73]			
AKM0			[0.54, 0.84]			

Notes: $N = 1,444$ (722 CZs $\times$ 2 time periods). Observations are weighted by the start of period CZ share of national population. All regressions include the full vector of baseline controls in ADH; that is those in Autor, Dorn, and Hanson (2013), table 3, column (6). 95% confidence intervals are reported in square brackets. Robust is the Eicker-White standard error. Cluster is the standard error that clusters CZs in the same state. AKM is the standard error in equation (37) with three-digit SIC clusters; AKM0 is the confidence interval with three-digit SIC clusters described in the last sentence of Section VA.

Table V. Chinese import competition and U.S. commuting zones

TABLE VI
ESTIMATION OF INVERSE LABOR SUPPLY ELASTICITY

Dependent variable:	First-stage L_i (1)	Reduced-form $\hat{\omega}_i$ (2)	2SLS $\hat{\omega}_i$ (3)
Panel A: Bartik IV, not leave-one-out estimator			
$\hat{\beta}$	0.90	0.73	0.80
Robust	[0.70, 1.10]	[0.54, 0.91]	[0.64, 0.97]
Cluster	[0.64, 1.16]	[0.47, 0.98]	[0.60, 1.01]
AKM	[0.65, 1.16]	[0.49, 0.96]	[0.62, 0.98]
AKM0	[0.61, 1.17]	[0.44, 0.96]	[0.59, 1.02]
Panel B: Bartik IV, leave-one-out estimator			
$\hat{\beta}$	0.87	0.71	0.82
Robust	[0.68, 1.06]	[0.53, 0.89]	[0.65, 0.98]
Cluster	[0.62, 1.12]	[0.46, 0.96]	[0.60, 1.03]
AKM (<i>leave-one-out</i>)	[0.59, 1.15]	[0.47, 0.94]	[0.61, 1.02]
AKM0 (<i>leave-one-out</i>)	[0.53, 1.15]	[0.42, 0.94]	[0.59, 1.09]
Panel C: ADH IV			
$\hat{\beta}$	-0.72	-0.48	0.67
Robust	[-1.04, -0.39]	[-0.80, -0.16]	[0.36, 0.98]
Cluster	[-0.93, -0.50]	[-0.78, -0.18]	[0.35, 0.99]
AKM	[-1.19, -0.24]	[-0.88, -0.07]	[0.27, 1.07]
AKM0	[-1.83, -0.35]	[-1.27, -0.10]	[0.18, 1.14]

Notes: $N = 1,444$ (722 CZs $\times$ 2 time periods). The variable L_i denotes the log change in the employment rate in CZ i . The variable $\hat{\omega}_i$ denotes the log change in mean weekly earnings. Observations are weighted by the start of period CZ share of national population. All regressions include the full vector of baseline controls in ADH; that is, those in Autor, Dorn, and Hanson (2013), table 3, column (6). 95% confidence intervals are in square brackets. Robust is the Eicker-White standard error; Cluster is the standard error that clusters CZs in the same state; AKM is the standard error in equation (37) with three-digit SIC clusters; AKM0 is the confidence interval with three-digit SIC clusters described in the last sentence of Section VA; AKM (*leave-one-out*) is the standard error in Section VC with three-digit SIC clusters; AKM0 (*leave-one-out*) is the confidence interval with three-digit SIC clusters described in Section VC.

Table VI. Inverse labor supply elasticity

Read: Table V applies the method to the Autor-Dorn-Hanson import competition design. AKM and AKM0 intervals are generally wider than robust or state-clustered intervals, particularly for 2SLS and reduced-form estimates. Table VI applies the method to Bartik-style labor-supply regressions. For the Bartik IV, AKM-type intervals are similar to conventional intervals; for the ADH IV, the new intervals widen substantially in the first-stage and reduced-form equations.

Economic message: The practical importance of AKM inference is application-specific. Some shift-share regressions are robust to the correction, while others are much less precise once the sectoral structure of the residuals is acknowledged.

8 Short summary

The paper argues that shift-share research designs require a special inference framework. The exposure-weighted regressor creates correlation across regions with similar shares, and the residual may contain unobserved shift-share shocks. Standard robust or geographic-clustered standard errors can therefore severely overreject. The authors propose AKM and AKM0 procedures that aggregate residuals through sector shares and use the shifter-level variation as the basis for inference. The method is especially useful when shifters are plausibly independent across many sectors or sector clusters.

9 Conclusion

Adão, Kolesár, and Morales provide one of the key modern foundations for inference in shift-share designs. The paper does not reject the use of Bartik-style designs; instead, it clarifies what must be true for inference to be valid.

The central takeaway is that the relevant correlation structure is not necessarily geographic. In shift-share designs, two regions can be statistically connected because they have similar industry composition. If residuals contain unobserved industry-level or group-level shocks, geographic clustering can miss the main source of dependence.

The paper’s recommended practice is to report confidence intervals that allow for a shift-share structure in the residuals. In applied work, this means asking whether the shifters are plausibly as good as randomly assigned, whether they are independent or clustered, and whether a few large shifters dominate the identifying variation.

The broader contribution is to make shift-share empirical work more transparent. The researcher must state where the randomness comes from: the shifters, the shares, or some other source. Once that source is explicit, inference should be built around it.